

RHUTAM MAHAJAN

College Park, MD

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EDUCATION

University of Maryland, College Park

M.S. in Data Science | GPA: 3.95/4.0

Aug 2025 – May 2027

College Park, MD

Savitribai Phule Pune University

B.E. in Computer Engineering, Honors in Data Science | GPA: 8.5/10

Dec 2021 – Jun 2025

Pune, India

TECHNICAL SKILLS

Languages & Core Libraries: Python, SQL, Pandas, NumPy, SciPy, scikit-learn

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, computer vision, NLP, few-shot and multimodal learning, time series forecasting

Generative AI & LLMs: OpenAI Agents SDK, Model Context Protocol (FastMCP), LangChain, LangGraph, ChromaDB, RAG, tool calling

Big Data, Cloud & Tools: Apache Spark, PySpark, Spark MLlib, Neo4j, MySQL, AWS (ECR, S3, App Runner), Docker, FastAPI, Git, Tableau, Power BI

EXPERIENCE

InternPe

AI/ML Intern

Feb 2024 – Mar 2024

Remote

- Developed four end-to-end ML projects across classification, regression, and neural nets, and deployed three as live Streamlit apps.
- Fixed train/test leakage in an IPL win model with a match-level split, not deliveries, for an honest 73% test accuracy.
- Engineered a 0.81 ROC-AUC diabetes screening model; handled invalid zero readings and tuned recall from 0.50 to 0.69.
- Cross-validated a car-price regressor to R^2 0.64; a Keras net matched a logistic baseline on cancer data at 0.99 AUC.

Gamaka AI

Junior Data Scientist

Dec 2023 – Mar 2024

Pune, India

- Designed normalized MySQL schemas of up to 12 tables with key integrity; cleaned missing values, duplicates, and outliers.
- Wrote analytical SQL with joins, grouping, and year-over-year logic to report revenue, profit, and EPS across seven companies.
- Implemented classification, regression, clustering, and time-series models that reached 99% on spam and 97% on digits; delivered Tableau/Power BI dashboards.

PROJECTS

Agentic AI Systems with MCP, RAG, and Evaluator Loops

Apr 2026

- Architected async multi-agent workflows across research, tools, memory, retrieval, and evaluation over a custom FastMCP server.
- Created a RAG chatbot with ChromaDB vector search, OpenAI generation, and Gemini-based evaluation.
- Orchestrated tool-calling loops so agents could plan a step, call a function, and revise the answer from evaluator feedback before replying.
- Refined the prompts and retrieval from the logged evaluator scores, which noticeably reduced hallucinated responses.

Spark Readmission Pipeline & Graph Data Engineering

Apr 2026

- Constructed a PySpark pipeline on 12,465 records; generated 16 features via Spark ML and mapped relationships in Neo4j.
- Wrangled raw admission records in Spark and resolved missing fields and skewed distributions before any modeling.
- Queried the Neo4j graph with Cypher to surface patient and provider relationships that the flat features missed.
- Trained a Spark MLlib Random Forest: 60.57% accuracy, 58.29% F1, and 0.636 AUC on held-out data.

Multimodal Medical Triage

Dec 2025 – Mar 2026

- Fused ResNet-18 lesion-image features with clinical text in a PyTorch model: 80.13% accuracy, 77.32% macro F1.
- Balanced the image and text branches during fusion so one modality would not drown out the other.
- Added a confidence threshold that sent low-certainty cases to manual review rather than a forced prediction.
- Shipped a FastAPI + React/TypeScript app, Dockerized and deployed to AWS ECR, App Runner, and S3 via GitHub Actions.

Plant Leaf Disease Multilevel Classification with Few-Shot Learning

Aug 2024 – May 2025

- Devised a two-level few-shot classifier on EfficientNet-B0 with Prototypical Networks that mapped 38 classes to 12.
- Sampled episodic support and query sets so the prototypical network learned in a genuine few-shot setting.
- Trained EfficientProtoNet on 240 images: 97.77% accuracy, 95.83% macro F1 on 13,135 held-out images.
- Published the work in IJRAR, an international peer-reviewed journal, Vol. 12, Issue 2, May 2025.